Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/03/2011
PI Name:	Britten, Richard Ph.D.		
Project Title:	Proteomic aided investigation of the mechanistic basis for HZE-induced cognitive impairment and the development of diagnostic biomarkers		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Brittera@evms.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	757-446-5038
Organization Name:	Eastern Virginia Medical School		
PI Address 1:	Radiation Oncology		
PI Address 2:	700 W Olney Rd		
PI Web Page:			
City:	Norfolk	**State:**	VA
Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received extension to 12/31/2011 per C. Guidry/JSC (Ed., 2/17/11)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Drake, Richard (Eastern Virginia Medical School) Lonart, Gyorgy (Eastern Virginia Medical School) Sanford, Larry (Eastern Virginia Medical School)		
Grant/Contract No.:	NNJ06HD89G		
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Task Description:	One of the main concerns that NASA has about the long-term health of the astronauts who go to Mars is the impact that cosmic radiation will have on brain function. Cosmic radiation consists of charged particles, known as Hze particles. The biological effects of Hze particles are dependent upon how much energy the particles deposit when they interact with tissue, which is referred to as LET. At present we only have superficial knowledge of how some high LET radiations inhibit brain function. Our current data suggest that brain function is inhibited due to radiation-induced death of brain cells and the inhibition of the brain's signaling pathways. However, the relative importance of each of these processes at different LETs is unclear. Such information is not only vital to the development of risk assessment models, but may provide an opportunity to develop medical treatments that can slow down or even stop the progression of radiation-induced brain impairment. The central objectives of our proposed studies address Research questions 29e, 29h and 29i outlined in the current Bioastronautic Roadmap. We specifically propose to: 1) identify the high-LET radiations that have the greatest impact on the ability of animals to recall imprinted memories and to learn new memory skills; 2) use PET imaging to identify the regions of the brain that are functionally inactivated when brain function is inhibited by Hze radiation; 3) establish the relative importance of radiation-induced cell death and signal malfunction in the inhibition of brain function after exposure to radiations with low, medium and high-LETs; 4) develop a blood-based assay that can be used to monitor the astronauts for early signs of radiation-induced memory function deficit. Specific Aim 1 will determine the dose required to inhibit four different memory functions in rats exposed to radiations with different LETs. The heads of rats will be exposed to graded doses of various LET radiations and the rats monitored over the following 6 months for signs of an impairment of their ability to retrieve memory and to learn new memory dependent tasks. Specific Aim 3 will determine the changes in the brain that lead to the loss of memory function in rats exposed to high-LET radiations. We shall determine the impact that radiation exposure has on the proliferation of brain cells by microscopic examination of brain slices. We shall also determine if radiation affects the ability of brain cells to release signaling proteins in response to known stimuli. Finally we shall use the new technology of proteomics, which can simultaneously measure the abundance of thousands of proteins, to identify all processes that are altered by radiation and are associated with the loss of brain function. Specific Aim 4 will use proteomics to establish the serum protein profiles of rats (who receive whole body radiation exposures) that go on to develop radiation-induced impairment of brain function. These proteins can be used as biomarkers with which to identify astronauts who have been exposed to radiation doses/energies that put them at risk of developing brain impairment. Our studies will establish the safe dose of cosmic radiation that astronauts can be exposed to without suffering from brain malfunction, which will be invaluable in deciding if the Mars mission should proceed, and if so, what additional precautions will be needed to protect the astronauts. Furthermore, our studies will provide considerable insight into how high LET radiation exposure leads to brain function impairment. These data will help develop treatments that can reduce the severity of radiation-induced brain impairment, e.g., special diets to reduce radiation-induced brain cell death, or drugs that can alter the brain's signaling pathways. Finally, our studies will identify biomarkers that can be used to identify and monitor astronauts who are at risk of developing radiation-induced brain impairment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of serum biomarkers for cognitive impairments has potential benefits in the diagnosis of neurological disorders.
Task Progress:	The majority of the work conducted this year has focused on obtaining sufficient data points to achieve statistical significance for the observations reported in Year 3 of the grant. We have now sufficient data points to submit manuscripts (one already under review and revision) on the Barnes maze and attentional set shifting data. However, progress was hindered by the initial decision not to run 56Fe in the Spring run at NSRL. This reporting period saw the publication of two papers, one relating to the impact that exposure to 56Fe had on the neurotransmission capability of hippocampal synaptosomes, the other reporting our identification of serum biomarkers of spatial memory impairment. We instigated a pilot study to determine if Aged rats had a similar susceptibility to HZE-induced neurocognitive impairment (Barnes Maze performance). Preliminary studies suggest that they are at least as susceptible and possible more susceptible that the young rats we have previously tested. Preliminary studies were also conducted that determined if exposure to 1.1 GeV Ti ions induced spatial memory impairments, and the data suggest that at <20 cGy there is measureable impairment of spatial memory performance. The major new developments during the reporting period was the progress made in identifying diagnostic biomarkers of neurocognitive impairment in the CSF of rats exposed to 1 GeV 56Fe. Our preliminary findings were presented by Dr. Singletary at ALAAS on the session dedicated to Space Veterinary Medicine. We have identified, although not fully verified, that two proteins are uniquely expressed in the CSF of rats that are resistant to HZE exposure and four proteins are uniquely expressed in rats that have developed HZE-induced spatial memory impairment. We also obtained CSF fluid from rats prior to irradiation to determine if there are prognostic biomarkers in the CSF that could be sued to predict the individual susceptibility to HZE-induced spatial memory impairment. Due to logistical constraints at EVMS these samples have not yet been run, but will be under the no-cost extension. Similarly we have made limited progress on the MALDI-MSI imaging due to the refitting of the Bruker machines used for this purpose. However, we have made some progress that should allow us to determine changes in the proteome, the lipoproteome and glycoproteome of the hippocampi from rats who have HZE-induced neurocognitive defects.

Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Britten RA, Mitchell S, Johnson AM, Singletary SJ, Keeney SK, Nyalwidhe JO, Karbassi ID, Lonart G, Sanford LD, Drake RR. "The identification of serum biomarkers of high-LET radiation exposure and biological sequelae." Health Phys. 2010 Feb;98(2):196-203. PMID: 20065683 , Feb-2010
Articles in Peer-reviewed Journals	Machida M, Lonart G, Britten RA. "Low (60 cGy) doses of (56)Fe HZE-particle radiation lead to a persistent reduction in the glutamatergic readily releasable pool in rat hippocampal synaptosomes." Radiation Research. 2010 Nov;174(5):618-23. Epub 2010 Aug 12. PubMed PMID: 20726706 , Nov-2010